Contact

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www.linkedin.com/in/ilyakipnis (LinkedIn)
www.QuantStratTrader.wordpress.co (Blog)
www.Github.com/IlyaKipnis (Portfolio)

Top Skills

Predictive Modeling
Financial Metrics
Quantitative Risk

Languages

Russian

Certifications

Intro to SQL for Data Science
Python Developer
Data Analyst With Python
Joining Data in PostgreSQL
Cohort 5 Fellow

Publications

Quantitative Trading With R
Momentum and Markowitz: A Golden Combination

Ilya Kipnis

Quant researcher, data scientist, Climatebase fellow, trader, machine learning professional. R instructor, Python, SQL
New York, New York, United States

Summary

I am a data science, machine learning, and systematic trading professional familiar with techniques like SVMs, random forests, xgboost, PCA, Monte Carlo simulation, and much more. I have a wealth of expertise in the R statistical programming language, and skill in Python.

I have implemented bespoke data science algorithms that were able to couple data extraction queries with efficient data transformation algorithms for fast dashboard visualization that generated tens of millions of dollars in revenue for large multinational corporations, created bespoke analytical solutions for smaller clients, and more.

A portfolio of my work is available at https://quantstrattrader.wordpress.com as it contains several years of open-source programming I have made freely available, and is voted one of the most helpful resources in statistics and data analysis by Quantocracy.

Lastly, I run my own systematic trading strategy that individuals can subscribe to at https://patreon.com/quantstrattrader . It is a trend following strategy in the volatility ETF space that has managed to minimize losses during choppy markets, and make substantial returns during both calm and crisis markets.

Experience

Quantstrat Trader

12 years 4 months

Volatility Trader

April 2017 - Present (9 years 3 months)

• Published original prototype on website in late 2014 on systematic trading research site in R.

• Iterated on volatility trading system trading long and short volatility ETPs.
• Implemented both an aggressive and conservative variant.
• Started trading in mid 2017 and dodged VIX spike on February 5, 2018
• Captured part of the long volatility upside during March 2020 Covid-19 market crisis.

Author
March 2014 - Present (12 years 4 months)

My blog and portfolio of data science, quantitative trading research, asset allocation, R programming, and more.

Open Source Financial Applications in R
Open Source Developer
June 2011 - Present (15 years 1 month)

• Implementing open-source software for the R language as it pertains to financial applications.
• Examples of such can be seen on my blog, https://www.quantStratTrader.wordpress.com , read worldwide by industry veterans
• Results on blog clearly communicated, with code written for results to be easily replicated and appended by end users.

The Carlyle Group
Quant/Systems Analyst
March 2024 - December 2024 (10 months)
Washington DC-Baltimore Area

• Implemented Quantitative Analytics to assess attractiveness of proposed Private Equity investments as compared to publicly traded companies by size, sector, etc.
• Refactored, implemented, and deployed robust reporting process for legacy asset allocation infrastructure to improve runtimes from multiple days to hours
• Implemented analytics to gauge viability of various 'blitzscaling' type startups and prospects of successful exits.
• Implemented methodologies for processing outliers in private equity investment analysis

Climatebase
Climatebase Fellow
March 2024 - May 2024 (3 months)

Climatebase Cohort 5 fellow--learned about the various efforts in climate to try and bring the temperature down.
Met and networked with some very interesting people eager to work in climate.
Learned about some interesting concepts such as circular economies, upcycling, nature-based solutions, and so on.
Participated in some lively climate-based discussions.

The Milwaukee Company
Quantitative Researcher
June 2021 - December 2023 (2 years 7 months)

Part time quantitative researcher for a private family office.
Wrote and improved a quantitative algorithm managing ~$100 million.
Researching signals to predict riskiness of downtrending or choppy markets as pertaining to systematic asset allocation.

Deutsche Bank
Application Developer
September 2021 - September 2022 (1 year 1 month)

• Implemented test cases to maintain code robustness on enterprise-level production code.
• Implemented analytics to track down data consistency errors, uncovering and fixing flaw in model forecasting input pipeline.
• Assisted and taught R programming as necessary.

Bank of America
Application Architect
August 2020 - July 2021 (1 year)

-Created reports for client accounts with bank, for both point-in-time, and historical reporting.
-Implemented python scripts for more efficient reporting and data analytics routines.
-Authored parts of future employee training handbook on how to use internal company software for smoother learning curves in the future.
-Designed components of a new API for trading reconciliation database.

Yewno, Inc.
Quantitative Analyst/Data Scientist
April 2018 - September 2018 (6 months)

Remote quantitative researcher/data scientist consultant.

• Modeled and reported effectiveness of AI-based signals.
• Created uniform infrastructure to perform comprehensive reporting of investment ideas.
• Modeled impact on turnover policies of proprietary strategies.
• Integrated SQL queries to obtain proprietary data from complex Yewno databases.
• Improved proprietary backtesting infrastructure to better reflect real-world investment constraints.
• Implemented portfolio optimization backtest in R on portfolio consisting of thousands of turnover periods and hundreds of assets to analyze impact of turnover reduction methodologies.
• Completed projects have gone onto be sold to outside clients.

BP
Front Office Developer/Quantitative Energy Analyst
October 2015 - July 2017 (1 year 10 months)

• Generated ~$100 million in revenue over the course of 22 months ($30 million in the first six months).
• Integrated and modified large SQL queries with RODBC.
• Implemented algorithms in R to process data with tens of millions of records regarding national gasoline pricing.
• Implemented data analysis algorithms in R to assist in visualization of planned and unplanned refinery outages.

Optimal Asset Management, Inc
Research Engineer
June 2013 - May 2014 (1 year)
Los Altos, CA

Reported to Vijay Vaidyanathan and Monty Joshi.

• Implemented R algorithms to calculate comprehensive suite of performance and risk metrics for smart beta portfolios.
• Constructed systematic, multi-strategy portfolios and reports for developed and emerging market smart beta products.
• Programmed software in R to create reports for $250M emerging market smart beta portfolio.
• Implemented R algorithms for reporting real-time performance of $250 million emerging market smart beta portfolio.

• Implemented R algorithms for portfolio reporting for smart beta product benchmark comparison.

Independent
Consultant
March 2010 - June 2013 (3 years 4 months)

Consulted in various quantitative capacities to several different firms. Consulting included construction of funnels and networks for analyzing start-ups and other early-stage private companies, speeding up analytical processes at a high-frequency firm by close to 50%, implementing processes to aid in discovery of statistical arbitrage pairs trading opportunities, as well as conducting and teaching data visualization in R.

Cheiron Trading
Quantitative Data Analyst Intern, Statistical Arbitrage Group
April 2011 - July 2011 (4 months)

Coded R full time in order to improve the data acquisition process, constantly improving the robustness of the data process to ensure proper data acquisition.

Analyzed historical trading results to improve trading strategies.

Lehigh University
Student
August 2005 - May 2009 (3 years 10 months)

Student at Lehigh University. Operations Research, statistics, more finance coursework than a finance major, financial engineering, and other coursework.

FIX Flyer
Support Desk Engineering
2009 - 2009 (less than a year)

Intern at FIX Flyer for George Kledaras.

Built documentation for FIX (Financial Information eXchange) protocol 5.0
Tested Flyer Online for bugs, an online trade settlement program.
Built test cases for FIX Flyer Ignition, a FIX protocol trade-validation simulation system in FIX Protocols 4.0, 4.2, 4.4, and 5.0.
Wrote a basic manual for using FIX Flyer Ignition for future office users and clients.

Assembled algorithms to translate messages across different FIX protocols in real-time using FIX Flyer's Daytona proprietary system.
Also suggested several improvements to the next iteration of FIX Flyer Daytona.

Watson Wyatt
Actuarial Intern
June 2008 - August 2008 (3 months)

Intern at Watson Wyatt. Worked in the retirement practice in Philadelphia. Checked paperwork and filled out government forms.

Drexel University
Lab Researcher
June 2007 - August 2007 (3 months)

Worked on the KIAPS (Kiaps Is A Plasma Squeegee) robot concept demonstrator for NASA

Education

Thinkful
Bootcamp Program, Data Science · (2020 - 2020)

Rutgers University
M.S., Statistics · (2009 - 2010)

Lehigh University
Info Systems Engineering, Operations Research, Financial Engineering, Economics · (2005 - 2009)

Cherry Hill East